Experiment No : 1

Title : String operations with and without pointers to arrays

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Program:

```
#include <stdio.h>
#include <string.h>

int main()
{
   char str[100], copy[100], rev[100], sub[100];
   int choice, i, j, len, pos, flag;

   printf("Enter a string: ");
   scanf("%s", str);

   do
   {
      printf("\n\n--- STRING OPERATIONS ---");
      printf("\n1. Copy");
      printf("\n2. Palindrome");
      printf("\n3. Reverse");
      printf("\n4. Substring");
      printf("\n5. Compare");
      printf("\n6. Exit");

      printf("\nEnter your choice: ");
      scanf("%d", &choice);

      len = strlen(str);

      switch(choice)
      {
```

```
case 1:
    strcpy(copy, str);
    printf("Copied string: %s", copy);
    break;

case 2:
    flag = 1;

    for(i = 0, j = len - 1; i < j; i++, j++)
    {
        if(str[i] != str[j])
        {
            flag = 0;
            break;
        }
    }

    if(flag)
        printf("The given string is a Palindrome");
    else
        printf("The given string is not a Palindrome");

    break;

case 3:
    for(i = 0; i < len; i++)
        rev[i] = str[len - 1 - i];

    rev[len] = '\0';

    printf("Reversed string: %s", rev);
    break;

case 4:
    printf("Enter starting position: ");
```

```
    scanf("%d", &pos);

    if(pos >= 0 && pos < len)
    {
        j = 0;

        for(i = pos; i < len; i++)
            sub[j++] = str[i];

        sub[j] = '\0';

        printf("Substring: %s", sub);
    }
    else
    {
        printf("Position out of range");
    }

    break;

case 5:
    {
        char str2[100];

        printf("Enter another string: ");
        scanf("%s", str2);

        if(strcmp(str, str2) == 0)
            printf("Both strings are equal");
        else if(strcmp(str, str2) < 0)
            printf("%s comes before %s", str, str2);
        else
            printf("%s comes after %s", str, str2);

        break;
```

```
            }

          case 6:
            printf("Exiting...");
            break;

          default:
            printf("Invalid choice!");
        }

    } while(choice != 6);

    return 0;
}
```

Output:

```
Output

Enter a string: Lion


--- STRING OPERATIONS ---
1. Copy
2. Palindrome
3. Reverse
4. Substring
5. Compare
6. Exit
Enter your choice: 1
Copied string: Lion
```

```
--- STRING OPERATIONS ---
1. Copy
2. Palindrome
3. Reverse
4. Substring
5. Compare
6. Exit
Enter your choice: 2
The given string is not a Palindrome
```

```
--- STRING OPERATIONS ---
1. Copy
2. Palindrome
3. Reverse
4. Substring
5. Compare
6. Exit
Enter your choice: 3
Reversed string: noiL
```

```
--- STRING OPERATIONS ---
1. Copy
2. Palindrome
3. Reverse
4. Substring
5. Compare
6. Exit
Enter your choice: 4
Enter starting position: 5
Position out of range
```

```
--- STRING OPERATIONS ---
1. Copy
2. Palindrome
3. Reverse
4. Substring
5. Compare
6. Exit
Enter your choice: 6
Exiting...
```